

# Wajih Habrah

Biomedical Engineer | Robotics, Digital Twin and Medical Technology

[wajih.habrah@hotmail.com](mailto:wajih.habrah@hotmail.com) | +46 737 000 738

Kapellgången 3, 41131 Gothenburg, Sweden

## EDUCATION

### Chalmers University of Technology

Gothenburg, Sweden

*Master's Degree in Biomedical Engineering*

*Sep. 2023 – Jun. 2026*

- **Thesis:** Robotic Phantom Knee and Digital Twin Platform for Early-Stage Exoskeleton Validation.
- **Selected coursework:** Deep Machine Learning, Applied Digital Health, Image Analysis, MRI, Rehabilitation Engineering, Modelling and Simulation in Biomedical Engineering, Applied Signal Processing, Advanced Topics in Biomedical Engineering.

### Damascus University

Damascus, Syria

*Bachelor's Degree in Biomedical Engineering*

*2007 – 2013*

- **Selected coursework:** Biomechanics, Biomedical Control, Automatic Control, Modelling and Simulation, Signal Processing, Medical Electronics & Bio-measurements, Medical Devices, Microprocessors, Artificial Extremities, Image Systems & Image Processing.

## WORK EXPERIENCE

### Biomedical Engineer (5 years)

Jan. 2018 – Dec. 2021 full-time; Jan. 2022 – Present hourly/consulting

*Southern hospital of Stockholm - Södersjukhuset*

*Stockholm, Sweden*

- Performed maintenance, calibration and troubleshooting of ICU systems, C-arms and critical diagnostic medical equipment.
- Developed and improved preventive maintenance procedures, including C-arm service routines, to increase operational reliability and safety.
- Produced and maintained technical documentation and supported clinical staff in equipment use, incident reporting and safety follow-ups.

## SELECTED PROJECTS

### Robotic Phantom Knee and Digital Twin | ROS 2, Unity, Python, C#, Raspberry Pi, Sensors, Actuators 2025 – 2026

- Designed and implemented a robotic knee testbed for early-stage exoskeleton validation, combining physical hardware with a real-time Unity digital twin.
- Developed bidirectional data flows between the physical system and the digital twin for real-time visualization, debugging and controller prototyping.
- Integrated variable-stiffness antagonistic actuation, pneumatic soft-tissue simulation and sensing from IMUs, pressure sensors, temperature sensors and encoders.
- Developed ROS 2 nodes for robotic motor control, pneumatic actuation, sensor integration, parameter management, logging and safety logic such as E-stop and air-release behavior.
- Built the Unity side of the system using ROS-TCP communication for interactive visualization and hardware-in-the-loop experimentation.

### Personal 3D and Interactive Systems R&D | Unity, Blender, Firebase, 3D Modelling

2019 – Present

- Explored interactive 3D systems using Unity and Blender, including asset workflows, real-time interaction and backend prototypes for dynamic configuration.
- Created 3D models and prepared assets for visualization, texture baking, low-poly modelling and 3D-printing workflows.

## TECHNICAL SKILLS

**Robotics & Control:** ROS 2, rclpy, real-time control, closed-loop control, cooperative actuation, variable-stiffness actuation, safety logic

**Digital Twins & Simulation:** Unity, ROS-TCP-Connector, hardware-in-the-loop visualization, Blender, 3D-printing preparation

**Programming:** Python, C#, C++, MATLAB

**Sensors & Hardware:** IMUs, pressure sensors, temperature sensors, encoders, Raspberry Pi, microcontrollers, pneumatic systems, medical equipment

**Tools:** Git, GitHub, GitLab, Unity Version Control, Microsoft Office

## LANGUAGES

---

**Arabic:** Native

**English:** Fluent

**Swedish:** Fluent

## ADDITIONAL

---

**Nationality:** Dual nationality, Syrian and Swedish

**Driving License:** Swedish class B, since 2017

**Interests:** Basketball, portrait drawing, sculpture, vehicle troubleshooting and repair

**References:** Available upon request